



EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test-1-II, Fall 2020 Semester

Course: CSE 207- Data Structures, Section-1
Instructor: Dr. Maheen Islam, Assistant Professor CSE Departmentment
Full Marks: 20
Time: 40 Minutes

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- 1 Ackerman's number, used in mathematical logic, can be calculated using the formula shown below. Write a recursive algorithm that calculates Ackerman's number. Verify your algorithm by using it to manually calculate the following test cases: 10 Marks
- a. Ackerman(2, 4),
 - b. Ackerman(3, 7),
 - c. Ackerman(0, 4), and
 - d. Ackerman(5, 0).

$$\text{Ackerman}(m, n) = \begin{cases} n + 1 & \text{if } m = 0 \\ \text{Ackerman}(m - 1, 1) & \text{if } n = 0 \text{ and } m > 0 \\ \text{Ackerman}(m - 1, \text{Ackerman}(m, n - 1)) & \text{otherwise} \end{cases}$$

- 2 What would be the value of queues Q1 and Q2, and stack S after the following algorithm segment: 10 Marks

```
1 S = createStack
2 Q1 = createQueue
3 Q2 = createQueue
4 enqueue (Q1, 5)
5 enqueue (Q1, 6)
6 enqueue (Q1, 9)
7 enqueue (Q1, 0)
8 enqueue (Q1, 7)
9 enqueue (Q1, 5)
10 enqueue (Q1, 0)
11 enqueue (Q1, 2)
12 enqueue (Q1, 6)
13 loop (not emptyQueue (Q1))
    1 dequeue (Q1, x)
    2 if (x == 0)
        1 z = 0
        2 loop (not emptyStack (S))
            1 popStack (S, y)
            2 z = z + y
        3 end loop
        4 enqueue (Q2, z)
    3 else
        1 pushStack (S, x)
    4 end if
14 end loop
```